Vishesh Gupta

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GITHUB // github.com/vishesh-gupta

SKILLS

Languages // Python, Go, C/C++, JavaScript/TypeScript
Frameworks & Libraries // gRPC, GraphQL, REST, React, Hugging Face, Lang Chain, NumPy, Pandas,
Tools // Docker, Kubernetes, AWS, GCP, Git, GitHub Actions, PostgreSQL, Protocol Buffers, Kafka Redis, Redpanda, RabbitMQ, IBM MQ

EXPERIENCE

STAFF SOFTWARE ENGINEER // ALPACA MARKETS — GLOBALLY REMOTE — SEPT 2026 – PRESENT

- Spearheaded infrastructure for 2 strategic partner launches: cut PII-redaction overhead by 86%; shipped FIX trading at very high volumes
- Engineered connectivity to 8+ market makers, clearing firms & regulators; kept 9 trading services up in shortages & sped up expansions
- Established DevOps standards: exposed 99.8% alert noise; authored 15+ RFCs, runbooks and policies, plus team charter

SENIOR SOFTWARE ENGINEER // ALPACA MARKETS — MAR 2025 – SEPT 2026

- Became OCC Infra SME in under 3 weeks, leading critical self-clearing integration under tight deadlines and adding resiliency & DR
- Architected trading interconnects and managed vendors: high availability, low-latency, fault-tolerant links with DR and network alerting
- Automated previously manual, communication-heavy processes into self-serve tooling, greatly improving developer velocity and QoL

SOFTWARE ENGINEER – INFRA LEAD // MONOCEROS — GRAND CAYMAN, KY — MAY 2024 – JAN 2025

- Independently built and managed Software and Hardware Infrastructure for 4 teams to handle market data flowing in the system
- Reliable infrastructure supported trading operations for multi-million dollars' worth of trading making huge impacts to all our traders
- Handled physical hardware for networking in multi office setup that ensures hardware reliability with best practices for high uptime

SOFTWARE ENGINEER // CLEAR STREET — NEW YORK, NY (REMOTE) — JAN 2021 – APR 2024

- Orchestrated integrations for Vault, Teleport, Artifactory, and Buf across engineering team, enforcing best practices and improving security and workflow efficiency
- Generated IaC automation tooling for Infrastructure Resource provisioning and faster CI/CD system with proper reviews and permission system. Modules to simplify resource creation in the IaC, reducing overhead by 40% in access request creation.
- Improved internal DNS routing using Transit Gateway and Route 53 Resolvers to handle DNS issues over the internal VPN network

PROJECTS

CRYPTOGRAPHIC ALGORITHMS // Basic encryption and decryption techniques

- Programmed basic concepts from cryptography such as Symmetric Key, Hashing, and Block Cipher
- Gained a strong understanding of Private and Public Key Cryptography and security techniques that help write secure software.

STATISTICAL MODELS & DISTRIBUTIONS // Collection of statistical models found in random research papers or articles

- Container to statistical models built from University courses or by reading about them in some research paper using R and Python
- Covers different distribution sets and graphical representations of datasets using Tableau and R libraries such as ggplot2

EDUCATION

UNDERGRADUATE // UNIVERSITY OF WATERLOO, WATERLOO, ONTARIO — SEPT 2016 – JAN 2021
Bachelor of Computer Science, Honors, Co-op // East Asian Studies Minor
Relevant Coursework: Data Structures & Algorithm, Operating Systems (C), Object Oriented Programming (C++), Numerical Computation (Python/Jupyter), Applied Cryptography

HOBBIES & INTERESTS

SPORTS & ADVENTURE ACTIVITIES // Badminton, Volleyball, Basketball, Baseball, Rock Climbing, Hiking, Skiing
INTERESTS // Anime, Music, Fine Dining, Networking, Travelling